



AICC: Air-combat intention-consistent coordination for cooperative UAV swarms

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ABSTRACT

Multi-UAV cooperative air combat faces incomplete information and long-horizon credit assignment challenges. In existing multi-agent reinforcement learning, the mapping between abstract coordination strategies and low-level control is often learned implicitly. This obscures interpretable high-level tactical semantics and can lead to a drift between coordination intent and action execution. We propose Air-Combat Intention-Consistent Coordination (AICC), a hierarchical multi-agent reinforcement learning framework. Within a centralized training and decentralized execution paradigm, AICC adopts an intention–maneuver hierarchy: a Manager emits low-frequency tactical intentions, while a Worker executes maneuvers conditioned on them using recurrent units to boost situational memory. To mitigate coordination mismatch, we introduce an intention prior mapping and enforce intention–action alignment via KL regularization and an intrinsic consistency reward. Furthermore, an intention-aware centralized Critic paired with a counterfactual baseline estimates marginal contributions to improve credit assignment. In 2v2 symmetric and 2v3 disadvantageous battles, AICC achieves more stable training and superior confrontation performance than benchmarks. Ablations confirm the benefits of hierarchical modeling, consistency supervision, and counterfactual credit assignment.

1. Introduction

Owing to their low cost, high mobility, rapid deployment, and strong environmental adaptability, unmanned aerial vehicles (UAVs) have evolved from “special-purpose platforms” into key aerial carriers supporting digitalization and intelligent operations across multiple industries [1]. In civilian applications, UAVs have been widely employed in emergency search and rescue and damage assessment, power-line and oil-gas pipeline inspection, agricultural plant protection and precision operations, surveying and remote sensing with three-dimensional reconstruction, traffic situation monitoring and urban governance, as well as logistics delivery, thereby significantly improving the timeliness, coverage, and operational safety of data acquisition [2]. In military and security applications, UAVs undertake a variety of missions, including intelligence reconnaissance, persistent surveillance, target indication, electronic countermeasures, and precision strikes, and their development is increasingly shifting from single-platform capability driven to system-level and swarm-level cooperative capability driven [3].

In the field of cooperative air-combat decision-making, representative methods can be broadly categorized into three complementary paradigms [4]. Expert-knowledge-driven rule-based systems employ finite-state machines, behavior trees, and scripted tactical

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libraries to realize formation maintenance, target assignment, and engagement procedures. Such methods exhibit strong interpretability and verifiability, but they are highly sensitive to rule completeness and manual design costs. Model-driven optimization and control methods, including mixed-integer programming, task allocation, model predictive control, as well as formation and consensus control, pursue approximately optimal cooperative behaviors under given dynamics and constraints. Although these methods provide solid theoretical support and controllability, they usually require high model accuracy and considerable computational resources. Game-theoretic and search-based methods utilize tools such as differential games, adversarial planning, and Monte Carlo tree search to perform opponent modeling and look-ahead search, thereby obtaining more robust action sequences. However, they still face challenges in sample efficiency [5] and generalization capability in high-dimensional state spaces and highly uncertain scenarios [6]. By comparison, reinforcement learning learns cooperative policies directly from interaction data, embedding temporal dependencies and nonlinear couplings into the policy and value functions, thus avoiding exhaustive rule enumeration and explicit global optimization. In particular, centralized training and decentralized execution improve data utilization and scalability [7]. Moreover, fine-grained advantage estimation further supports more reliable learning signals [8]. In addition, credit assignment mechanisms help distinguish individual contributions in multi-agent cooperation, thereby improving learning efficiency [9]. Overall, these factors enhance the upper bound of coordination performance in complex tactical spaces [10].

In multi-agent air-combat decision-making, maintaining strict consistency between high-level tactical intentions and low-level action execution remains a significant challenge [11]. While various approaches attempt to realize cooperation through optimization, game search, or standard reinforcement learning, many flat MARL architectures learn the mapping from observations to actions implicitly. Consequently, abstract tactical semantics are often entangled within the neural network's latent space, making it difficult to extract explicit and interpretable high-level intentions. Without an explicit hierarchical structure, the understanding and execution of coordinated tactics by different agents may gradually diverge. In practical situations, information asymmetry, sensor delay, and missing observations further introduce errors into the inference of implicit intentions by flat policies, making it challenging for agents to stably progress along a shared tactical trajectory.

To address these limitations, this paper proposes AICC framework. Rather than combining hierarchical policies, centralized critics, and regularization in a loosely coupled manner, AICC is built upon a unified intention-consistency learning principle. In this framework, tactical intention is explicitly represented and linked to low-level decision-making through a semantically grounded prior, enabling agents to align their behaviors with shared high-level objectives. Meanwhile, both policy learning and multi-agent credit assignment are guided by this intention-consistency constraint, forming a tightly coupled mechanism between high-level intention modeling and low-level execution. This design leads to more interpretable, stable, and coordinated behaviors, and distinguishes AICC from existing approaches that treat these components independently.

The main contributions are summarized as follows:

- **Intention-consistent coordination as a learning principle.** We explicitly model tactical intention as a latent variable and enforce consistency between intention and action, moving beyond implicit coordination in standard MARL.
- **Prior-guided intention–action alignment.** We introduce an intention prior mapping that injects domain knowledge into policy learning, together with KL regularization and an intrinsic consistency reward to align learned behaviors with tactical semantics.
- **Intention-aware counterfactual credit assignment.** We extend centralized critic learning by conditioning on intention variables, enabling more precise credit assignment in partially observable multi-agent environments.

The rest of this paper is organized as follows. Section 2 reviews the related work on UAV air-combat decision-making and multi-agent reinforcement learning. Section 3 presents the dynamical model, engagement determination, and the formal problem definition of the air-combat scenario as an incomplete-information dynamic game with private types. Section 4 introduces the modeling of the proposed AICC framework for intention-consistent cooperation, including the hierarchical intention–maneuver space, observation and state representation, as well as the design of the composite reward and credit assignment mechanism. Section 5 presents the deep-reinforcement-learning-based solution method for AICC cooperative policy learning, including the hierarchical policy network, the type-aware counterfactual Critic, and the consistency-supervised optimization objective. Section 6 reports simulation comparisons and ablation studies in multiple confrontation scenarios. Finally, Section 7 concludes this paper and discusses future work.

2. Related work

2.1. UAV air-combat decision-making

Against the backdrop of the continuous evolution of informationized warfare, the leap in battlefield situation awareness capability has significantly compressed the time window for operational decision-making, thereby imposing higher requirements on both real-time performance and reliability. In the study of autonomous maneuver decision-making for air combat, modeling and solving based on differential games has constituted an important technical route. Zhong et al. [12] proposed a differential-game-driven autonomous maneuver decision-making method, in which the originally complex bilateral extremum problem was transformed into a unilateral extremum form, thereby reducing the computational burden of joint prediction of future states of both sides while improving the real-time performance of online decision-making. Park et al. [13] developed a close-range air-combat maneuver decision-making algorithm in a continuous dynamic air-combat scenario, where a hierarchical decision-making structure and a payoff function matrix were designed to achieve efficient strategy selection through the computation of corresponding scores. On this basis, Lee et al. [14] further enhanced the representation and analysis capability for the complexity of opponent maneuver actions, and combined payoff-function evaluation with opponent maneuver analysis to optimize the solution of friendly maneuver actions, thus improving the

adaptability of the strategy to adversarial behaviors. Xu et al. [15] introduced a continuously differentiable air-combat situation function into the differential-game modeling of aircraft confrontation, and adopted a semi-direct method for numerical solution, thereby improving both the speed and accuracy of air-combat strategy generation while maintaining solution stability. While these methods provide strong theoretical grounding and real-time decision capabilities, they primarily operate at the level of maneuver optimization and payoff evaluation, where high-level tactical intentions are not explicitly represented but implicitly embedded in model design and objective functions, limiting their interpretability and flexibility in cooperative scenarios.

Meanwhile, the development of deep reinforcement learning has provided a more data-driven and end-to-end optimized toolset for air-combat decision-making. Kurniawan et al. [16] introduced the Actor–Critic framework into the air-combat confrontation task, where the learning process was regulated through reward-structure design to obtain effective tactical strategies. Jing et al. [17] realized UAV maneuver decision-making by improving the deep deterministic policy gradient (DDPG) algorithm, and enhanced the realism and task relevance of the simulation environment from the perspective of energy-based air-combat modeling. Hu et al. [18] proposed an improved deep Q -learning method for air-combat decision-making, and incorporated long short-term memory (LSTM) units to enhance temporal situation representation, enabling the agent to more effectively exploit its own advantages to threaten the target while avoiding enemy threats. To address the “curse of dimensionality” that often arises in traditional maneuver decision-making in high-dimensional continuous spaces, Yang et al. [19] employed methods such as deep deterministic policy gradient to directly output actions in continuous spaces so as to improve scalability. However, most reinforcement learning-based approaches focus on reward-driven policy optimization, where tactical intentions are implicitly encoded into reward functions or observations rather than explicitly modeled. As a result, the learned policies often lack semantic interpretability and controllability, especially in multi-agent cooperative settings where consistent high-level intention understanding is critical.

2.2. Multi-agent reinforcement learning

Centralized training and decentralized execution has become a mainstream paradigm in multi-agent reinforcement learning (MAREL). The core idea is to introduce globally available information during the training stage so as to improve the identifiability of learning signals and alleviate the non-stationarity caused by parallel policy updates. During deployment, however, the decentralized execution constraint is strictly enforced, such that each agent makes decisions independently based only on its own local observation, thereby maintaining the scalability and flexibility of the system without introducing additional online communication or information dependence [20]. In this way, the centralized training and decentralized execution (CTDE) achieves an effective utilization of global information that is available during training but unavailable during execution, and provides a favorable balance among stability, efficiency, and deployability at the methodological level. However, such frameworks typically model coordination implicitly through shared value functions or training signals, without explicitly introducing structured representations of high-level intentions.

Existing CTDE methods mainly inject global information into the learning process through two technical routes. The first is centralized value evaluation, in which a value function conditioned on global information is constructed during training to reduce the variance of gradient estimation, stabilize policy iteration, and improve credit assignment. During execution, decentralized policy networks are still adopted, thereby avoiding direct dependence on global information and introducing no extra communication overhead. Representative studies such as MAPG [21] and MADDPG have shown that a centralized critic can significantly improve training stability and final performance in cooperative tasks. Complementary to this line, value function decomposition characterizes the relationship between team value and individual contributions in a structured manner by representing the global action-value function as a combination of individual utilities, thereby supporting decentralized greedy decision-making and enhancing interpretability. VDN [22] employs additive decomposition to approximate the global Q -function, which substantially simplifies team-value modeling. Subsequent methods such as QMIX [23] and FOP [24] further improve the expressive capability of the mixing function, and strengthen the compatibility between locally optimal selection and global value improvement through consistency constraints such as individual-global-max (IGM) and individual-global-optimality (IGO).

Although CTDE exhibits clear advantages, its scalability in large-scale systems is still limited by the growing complexity brought by centralized modeling. As the number of agents increases, the dimensions of the states and actions that must be processed by the centralized Critic expand rapidly, leading to a significant increase in training cost. Meanwhile, in shared or coupled value learning, individually suboptimal behaviors may interfere with joint optimization through estimation bias and misattribution, thereby reducing cooperation efficiency and hindering the attainment of the global optimum. Therefore, how to suppress the scale effect and improve global consistency through more effective structural design and coordination mechanisms, while satisfying the decentralized execution constraint, remains a key issue for the further development of CTDE [25]. In addition, most existing methods focus on improving credit assignment or value factorization, while treating policy regularization primarily as a generic tool for stabilizing training. As a result, the relationship between high-level decision abstractions and low-level action selection is often not explicitly modeled, limiting the interpretability and controllability of learned cooperative behaviors.

3. Problem formulation

3.1. UAV dynamics model

In the multi-UAV cooperative air-combat scenario, each UAV is modeled as a point mass moving in three-dimensional space. A ground inertial coordinate system $\Sigma_g (O_g - X_g Y_g Z_g)$ is established. For the i -th UAV, the position vector and velocity vector are denoted by $\mathbf{p}_i = [x_i, y_i, z_i]^T$ and $\mathbf{v}_i = [\dot{x}_i, \dot{y}_i, \dot{z}_i]^T$, respectively. Its motion state vector is defined as $\mathbf{s}_i = [x_i, y_i, z_i, v_i, \theta_i, \psi_i]^T$, where v_i denotes the speed, θ_i denotes the flight-path pitch angle, and ψ_i denotes the flight-path yaw angle.

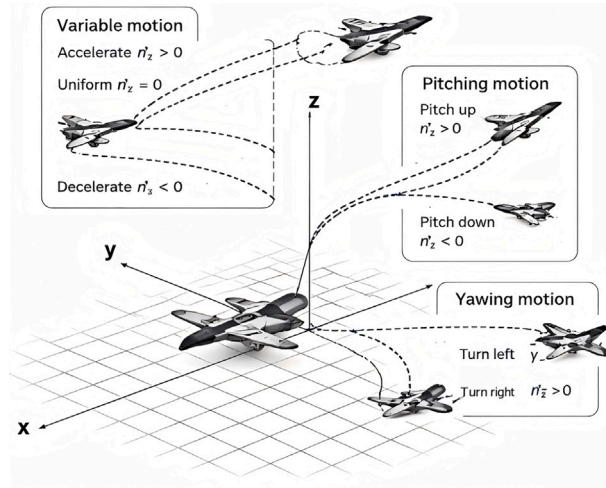


Fig. 1. UAV dynamics model.

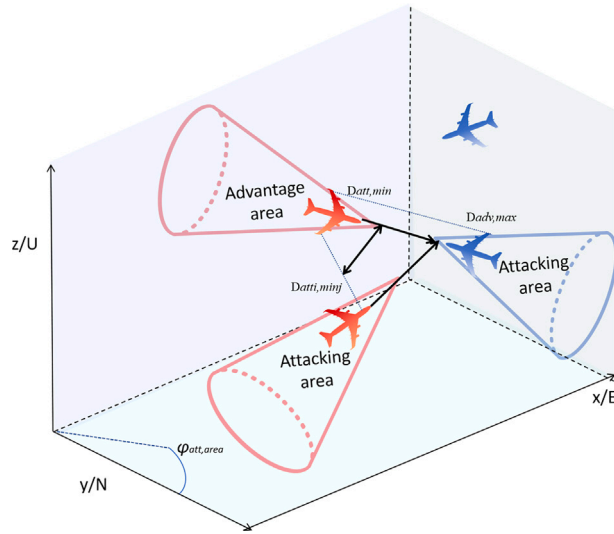


Fig. 2. Air-combat situation.

The maneuver of the UAV is controlled by the tangential overload $n_{x,i}$, the normal overload $n_{z,i}$, and the roll angle γ_i . The kinematic equations are given by

$$\begin{cases} \dot{x}_i = v_i \cos \theta_i \cos \psi_i, \\ \dot{y}_i = v_i \cos \theta_i \sin \psi_i, \\ \dot{z}_i = v_i \sin \theta_i, \\ \dot{v}_i = g(n_{x,i} - \sin \theta_i), \\ \dot{\theta}_i = \frac{g}{v_i}(n_{z,i} \cos \gamma_i - \cos \theta_i), \\ \dot{\psi}_i = \frac{g n_{z,i} \sin \gamma_i}{v_i \cos \theta_i}, \end{cases} \quad (1)$$

where g denotes the gravitational acceleration. To accommodate reinforcement learning algorithms with a discrete action space, the continuous control inputs are discretized into an action set containing M basic tactical maneuvers, denoted by $\mathcal{A}_{\text{low}} = \{a^1, \dots, a^M\}$. Within each decision interval ΔT , the agent selects one action $a_t \in \mathcal{A}_{\text{low}}$ and keeps executing it, while the state is updated by integrating the above equations. Meanwhile, all maneuvers are subject to physical constraints, including $v_i \in [v_{\min}, v_{\max}]$ and $n_{z,i} \in [n_{z,\min}, n_{z,\max}]$, as shown in Fig. 1.

Fig. 2 illustrates the simulated air-combat situation, where the confrontation between the red-side UAV set Ω_r and the blue-side UAV set Ω_b is considered. The outcome of the air combat depends on both the geometric situation and weapon effectiveness.

The relative distance vector between UAV i and target j is defined as $\mathbf{D}_{ij} = \mathbf{p}_j - \mathbf{p}_i$. The key attacking angle (AA) and escape angle (ATA) are defined as

$$\varphi_{\text{att},ij} = \arccos \frac{\mathbf{v}_i \cdot \mathbf{D}_{ij}}{\|\mathbf{v}_i\| \|\mathbf{D}_{ij}\|}, \quad \varphi_{\text{esp},ij} = \arccos \frac{\mathbf{v}_j \cdot \mathbf{D}_{ij}}{\|\mathbf{v}_j\| \|\mathbf{D}_{ij}\|}. \quad (2)$$

If and only if the following missile no-escape-zone conditions are satisfied, the red UAV i is regarded as having an effective attack capability against the blue UAV j :

$$\mathbb{I}_{\text{fire}}(i, j) = \begin{cases} 1, & \text{if } \|\mathbf{D}_{ij}\| \in [D_{\text{att},\min}, D_{\text{att},\max}] \\ & \wedge \varphi_{\text{att},ij} \leq \varphi_{\text{att},\max} \\ & \wedge \varphi_{\text{esp},ij} \leq \varphi_{\text{esp},\max}, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

In addition, to accelerate training convergence, the concepts of the attack area and the advantage area are introduced by following [26]. Specifically, dense tactical guidance rewards are provided by relaxing the angle thresholds ($\varphi_{\text{att},\text{area}}, \varphi_{\text{esp},\text{area}}$).

A probabilistic piecewise damage model is adopted. If the attack judgment is successful, the target health value B_j is reduced by $\Delta B(p_{\text{hit}})$, where p_{hit} denotes the stochastic hit probability. When $B_j \leq 0$, or when collision, ground impact, or boundary violation occurs, the target is judged as destroyed and removed from the scenario.

3.2. Partially observable stochastic game formulation

Based on the kinematic constraints, the multi-UAV cooperative air-combat problem is formulated as a partially observable stochastic game (POSG), which captures decentralized decision-making under partial observability and adversarial interaction. The game is described by the tuple

$$G = \langle \mathcal{N}, \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{O}, \mathcal{R}, \gamma \rangle, \quad (4)$$

where each component is specified as follows:

- **Agents ($\mathcal{N}$):** Let $\mathcal{N} = \Omega_b \cup \Omega_r$ denote the set of all UAVs participating in the game, where Ω_b represents the controllable blue-side UAV formation and Ω_r represents the red-side UAV formation with adversarial behaviors.
- **Global state space ($\mathcal{S}$):** The global state $s_t \in \mathcal{S}$ describes the physical configuration of the battlefield at time t , including the kinematic states of all UAVs, i.e.,

$$s_t = \{s_{i,t} \mid \forall i \in \mathcal{N}\}. \quad (5)$$

The global state governs environment evolution but is not directly accessible to any single UAV during execution.

- **Joint action space ($\mathcal{A}$):** The joint action space is defined as the Cartesian product of the low-level maneuver spaces of all UAVs, namely,

$$\mathcal{A} = \prod_{i \in \mathcal{N}} \mathcal{A}_{\text{low}}. \quad (6)$$

At each decision step, UAV i selects a maneuver command $a_{i,t} \in \mathcal{A}_{\text{low}}$.

- **State transition function ($\mathcal{T}$):** The transition function $\mathcal{T} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ describes the dynamical evolution of the environment. Given the current state s_t and joint action $\mathbf{a}_t$, the next state s_{t+1} is obtained by numerically integrating the UAV dynamics in Eq. (1).
- **Local observation function ($\mathcal{O}$):** Due to sensing and communication limitations, each UAV i receives only a local observation $o_{i,t} \in \mathcal{O}_i$. The observation contains the self-state and partial relative information of detectable entities, rather than the complete global state.
- **Reward function ($\mathcal{R}$):** The environment provides task-related feedback $r_{i,t}^{\text{ext}} = \mathcal{R}(s_t, \mathbf{a}_t)$, which reflects combat outcomes such as attack effectiveness, survival, and destruction events. The intention-related intrinsic reward used in AICC is introduced later as part of the learning framework in Section 4.3.

Under this formulation, the blue-side UAV formation seeks a decentralized policy π_{Ω_b} that maximizes the expected discounted cumulative return against the adversarial policy of the red side:

$$J(\pi_{\Omega_b}) = \mathbb{E}_{s_0 \sim \rho_0, \mathbf{a}_t \sim \pi, s_{t+1} \sim \mathcal{T}} \left[\sum_{t=0}^T \gamma^t \sum_{i \in \Omega_b} r_{i,t}^{\text{ext}} \right]. \quad (7)$$

On top of the above POSG, the proposed AICC framework introduces a discrete intention variable $z_{i,t}$ at the policy level, which is generated by a high-level Manager from the local observation history and then used to condition low-level maneuver selection. Therefore, $z_{i,t}$ is treated in this work as an internal latent decision variable for hierarchical control, rather than as an exogenous private type in the strict Bayesian-game sense. Correspondingly, the current implementation does not maintain an explicit Bayesian belief distribution $b_i(z_{-i} \mid h_{i,t})$ over teammates' intentions. Instead, coordination under partial observability is learned implicitly through recurrent observation encoding within the centralized-training-and-decentralized-execution framework.

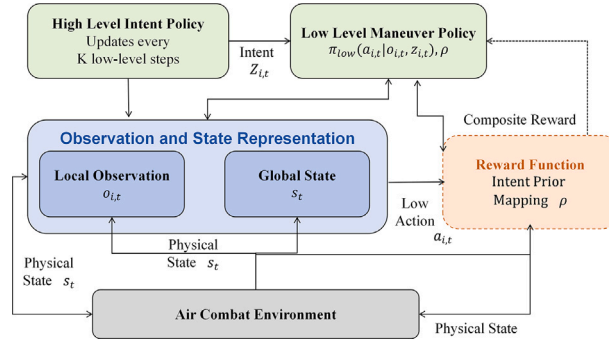


Fig. 3. Overall architecture of the AICC framework.

4. AICC framework modeling for intention-consistent coordination

To address the challenges of unobservable intentions and difficult credit assignment in multi-UAV air-combat games, this section presents the core modeling of the AICC framework in detail. As shown in Fig. 3, the proposed framework follows the centralized training and decentralized execution paradigm. By constructing a hierarchical decision-making structure, differentiated observation representations, and a composite reward mechanism, it aims to resolve the coordination challenges in complex dynamic environments.

4.1. Hierarchical action space and intention modeling

To enable efficient policy learning in continuous state spaces while endowing UAV behaviors with interpretability, the decision space is decoupled into a two-level “intention–maneuver” structure. The low-level controller is mainly responsible for translating concrete tactical commands into physical maneuvers satisfying dynamical constraints. Considering the high dimensionality of the continuous control space, the control vector of the i -th UAV, denoted by $u_i = [n_{x,i}, n_{z,i}, \gamma_i]^T$, is discretized into a finite basic maneuver library $\mathcal{A}_{\text{low}} = \{a^1, \dots, a^M\}$. This set covers M typical air-combat maneuvers, such as rapid acceleration, sharp turning, climbing, and diving, where each action corresponds to a fixed set of overload and roll commands. At decision time t , the low-level policy network of UAV i outputs an action $a_{i,t} \in \mathcal{A}_{\text{low}}$, which is then maintained over the control interval ΔT to drive the evolution of the dynamical equations. These explicit maneuver actions constitute observable behavioral cues in the environment and provide the physical basis on which other UAVs adapt their decisions under partial observability.

Above the low-level maneuver layer, a high-level tactical intention space $\mathcal{Z}$ is introduced to represent the tactical role and long-term plan of each UAV. For a blue-side UAV $i \in \Omega_b$, its high-level intention variable $z_i \in \mathcal{Z}$ takes values from the discrete set {primary attack, cover, positioning, evasion}. Unlike the high-frequency switching of low-level actions, the high-level intention exhibits clear temporal persistence. Specifically, the intention update interval is set to K control steps, i.e., $z_{i,t}$ is sampled and updated according to the global situation only when $t \bmod K = 0$, and remains fixed over the interval $[t, t + K)$. This hierarchical design decomposes the complex game into two subproblems: the top-level policy is responsible for macroscopic tactical planning, whereas the bottom-level policy handles microscopic maneuver execution. Here, $z_{i,t}$ is an internal latent intention variable generated by the high-level Manager for hierarchical control, rather than a predefined private type of the environment.

To associate the latent intention z with the explicit action a , an intention prior mapping $\rho : \mathcal{Z} \times \mathcal{S} \rightarrow \Delta(\mathcal{A}_{\text{low}})$ is defined. This mapping describes the ideal probability distribution over basic maneuver actions under a specific intention z . For example, when the intention is “primary attack,” maneuver actions pointing toward the enemy should have higher probability mass. The intention prior $\rho(a | s, z)$ is introduced as a conditional distribution over actions given the state s and the intention variable z . It links high-level intention to concrete maneuver execution, so that different intentions correspond to distinguishable action tendencies. In implementation, $\rho(a | s, z)$ is modeled by a differentiable function approximator $\rho_\psi(a | s, z)$ and learned together with the policy. Rather than being predefined, it is fitted to the action distribution generated by the current policy under the same intention. The following objective is used:

$$\mathcal{L}_\rho = \mathbb{E}_{(s,z,a) \sim D} [-\log \rho_\psi(a | s, z)], \quad (8)$$

where a is sampled from $\pi_\theta(\cdot | s, z)$. This allows ρ_ψ to gradually capture the typical behavioral patterns associated with each intention directly from interaction data. To keep the learning process stable, ρ_ψ is updated more conservatively, e.g., by using a slower update rate or a target network.

In the AICC framework, ρ serves not only as prior knowledge but also as the reference for intention-consistency supervision. The learned low-level policy $\pi_{\text{low}}(a_{i,t} | o_{i,t}, z_{i,t})$ is required to statistically approach this prior distribution by minimizing the KL divergence between them:

$$\min D_{\text{KL}}(\pi_{\text{low}}(\cdot | o_{i,t}, z_{i,t}) \| \rho(\cdot | s_{i,t}, z_{i,t})). \quad (9)$$

4.2. Observation space and state representation

To support intention inference and cooperative decision-making under incomplete information, the framework strictly distinguishes between the local observation of each UAV and the global state of the environment, while introducing relative geometric operators at the feature-engineering stage. For any pair of UAVs, the system first computes the normalized relative distance D_{hk} , the line-of-sight angles θ_{hk} and ψ_{hk} , as well as the velocity difference Δv_{hk} , which are used as the basic input features of the neural network.

During execution, UAV i can access only its local observation o_i^l . The observation vector $o_i = [o_i^{\text{self}}, o_i^{\text{friends}}, o_i^{\text{enemies}}]^\top$ contains its self-state and the kinematic information of entities within its field of view. Crucially, however, it strictly masks the high-level intentions z_j of friendly UAVs j . This information masking prevents the Actor network from relying on explicit intention communication. Instead, each agent makes decisions based on teammates' observable behaviors and the current relative situation. In the present implementation, such adaptation is achieved implicitly through recurrent encoding of local observation histories, rather than through an explicit belief update module over teammates' intentions.

In contrast, during centralized training, the Critic network is granted a global view to address the multi-agent credit assignment problem. The global state S_{global}^t aggregates the physical states s_i^{phys} of all UAVs together with their intention variables z_i :

$$S_{\text{global}} = \left[\bigcup_{i \in \mathcal{N}} s_i^{\text{phys}}, \bigcup_{i \in \mathcal{N}} z_i \right]. \quad (10)$$

Including the joint intention z in the Critic input improves value evaluation during centralized training, because it allows the Critic to distinguish different high-level coordination modes that may lead to different low-level action consequences under the same physical state. This design follows the centralized-training-and-decentralized-execution paradigm and does not imply that agents explicitly observe teammates' intentions during execution.

4.3. Reward function design

To address the problems of sparse returns and difficult credit assignment in cooperative air combat, a composite reward system consisting of external task rewards and intrinsic consistency rewards is designed. The overall return $r_{i,t}$ is composed of the external task reward ($r_{i,t}^{\text{shape}} + r_{i,t}^{\text{event}}$) and the weighted intrinsic consistency reward ($\beta \cdot r_{i,t}^{\text{consist}}$):

$$r_{i,t} = \underbrace{r_{i,t}^{\text{shape}} + r_{i,t}^{\text{event}}}_{\text{External Task Reward}} + \underbrace{\beta \cdot r_{i,t}^{\text{consist}}}_{\text{Intrinsic Reward}}. \quad (11)$$

The external task reward directly reflects the win-loss logic of the air-combat game. Among its components, the situation-shaping reward $r_{i,t}^{\text{shape}}$ is intended to accelerate exploration in the early stage of training. To this end, a potential function based on geometric situation is constructed by jointly considering distance and angle advantages, such that the reward of a UAV depends on its suppression level against the most threatening enemy target and its own risk of being locked. The exact functional form of the situation-shaping reward is defined as:

$$r_{i,t}^{\text{shape}} = \omega_d \cdot \left(1 - \frac{d_{i,\text{enemy}}}{D_{\text{max}}} \right) + \omega_a \cdot \frac{1}{2} (1 + \cos(\phi_{i,\text{enemy}})), \quad (12)$$

where $d_{i,\text{enemy}}$ represents the distance from UAV i to its most threatening enemy, D_{max} is the maximum radar detection range, and $\phi_{i,\text{enemy}}$ is the antenna train angle toward that enemy. The predefined weight coefficients are set to $\omega_d = 0.5$ and $\omega_a = 0.5$. Meanwhile, the sparse event reward $r_{i,t}^{\text{event}}$ assigns discrete rewards and penalties to critical tactical events, such as destroying an enemy UAV (+10, shared by all teammates to promote cooperation), self-destruction (−10), and effective locking (+0.5), so as to explicitly define the final task objective.

The intrinsic consistency reward $r_{i,t}^{\text{consist}}$ serves as the core mechanism for establishing credible signaling. To penalize behaviors with inconsistent intention and action, this reward is defined as the log-likelihood of the current action under the prior intention distribution:

$$r_{i,t}^{\text{consist}} = \log \rho(a_{i,t} | s_{i,t}, z_{i,t}). \quad (13)$$

This term acts as a soft constraint: if UAV i claims to execute a certain intention but takes an action with extremely low prior probability under that intention, it will receive a negative penalty. This drives the action and intention to remain statistically consistent, enabling teammates to effectively infer its intention from observations. To balance the main task objective and the auxiliary intention consistency, the weight for the intrinsic reward in Eq. (11) is empirically set to $\beta = 0.05$. It is crucial to emphasize that the computation of the intrinsic consistency reward $r_{i,t}^{\text{consist}}$ relies on the global state $s_{i,t}$, which is strictly confined to the Centralized Training phase. During training, this reward serves as an auxiliary signal to evaluate policies and update network weights via the centralized critic. However, during the Decentralized Execution phase, the computation of any reward signals is entirely bypassed. The worker UAVs generate their actions relying solely on their local observations $o_{i,t}$ and the assigned intentions $z_{i,t}$. Therefore, our framework strictly preserves the premise of decentralized execution without assuming any access to the global state at test time.

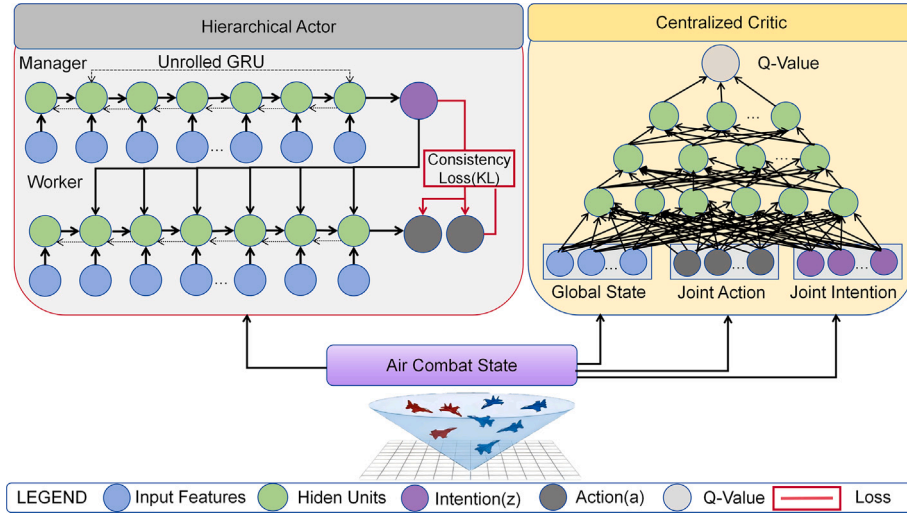


Fig. 4. Architecture of the AICC cooperative decision-making model.

In addition, to address the “free-rider” problem caused by directly summing team rewards, a counterfactual baseline is further constructed using global state information:

$$A_i(s_t, \mathbf{a}_t, \mathbf{z}_t) = Q(s_t, \mathbf{a}_t, \mathbf{z}_t) - \sum_{a'_i} \pi_i(a'_i | o_{i,t}, z_{i,t}) Q(s_t, (a'_i, \mathbf{a}_{-i,t}), z_t). \quad (14)$$

By computing the advantage of the specific action of UAV i relative to the expected performance of its average policy under the given intention, the team-level global return can be accurately decomposed into the marginal contribution of each individual UAV, thereby enabling precise gradient propagation during cooperative optimization.

5. Deep-reinforcement-learning-based solution for AICC cooperative policy learning

To address the partial observability and highly dynamic adversarial characteristics of multi-UAV air-combat games, the AICC framework is proposed. As shown in Fig. 4, the framework follows the centralized training and decentralized execution paradigm and integrates recurrent-neural-network-based hierarchical reinforcement learning with a type-aware multi-agent Actor–Critic architecture. Through explicit intention modeling and strict consistency supervision, the high-dimensional joint policy space is decoupled into two subproblems, namely long-horizon tactical decision-making and high-frequency maneuver execution, thereby improving long-horizon cooperative policy learning while preserving policy interpretability.

5.1. Hierarchical policy architecture based on recurrent neural networks

The air-combat decision-making process is inherently non-Markovian, since a single-step local observation is often insufficient to characterize the evolution of the confrontation situation, and effective decisions depend on accumulated memory of historical maneuvers and interaction outcomes. Meanwhile, air-combat strategies naturally require temporal abstraction from the tactical level to the maneuver level. To this end, AICC constructs a two-level hierarchical policy structure with embedded gated recurrent units for each UAV i , so as to explicitly capture both long- and short-term temporal dependencies and to decouple macroscopic tactical planning from microscopic action execution.

At the top of the hierarchy is the intention generation network, whose policy distribution is parameterized by μ_ϕ . The Manager is responsible for macroscopic tactical planning: it extracts higher-level tactical semantics from the local observation sequence and outputs discrete tactical intentions for the low-level controller to execute. To embody the hierarchical control principle of “low-frequency high-level decision-making and high-frequency low-level execution,” the Manager operates at a lower temporal frequency and updates the intention only once every K control steps. Specifically, to cope with partial observability, the Manager uses a GRU to aggregate historical information. At decision time t (when $t \bmod K = 0$), the Manager receives the current local observation $o_{i,t}$ and updates the hidden state $h_{i,t}^{\text{mgr}}$, after which a discrete intention $z_{i,t}$ is sampled from the latent space $\mathcal{Z}$:

$$h_{i,t}^{\text{mgr}} = \text{GRU}_{\text{mgr}}(o_{i,t}, h_{i,t-1}^{\text{mgr}}), \quad z_{i,t} \sim \mu_\phi(\cdot | h_{i,t}^{\text{mgr}}). \quad (15)$$

The intention $z_{i,t}$ is a latent high-level command generated by the Manager from the local observation history, rather than a predefined private type of the environment. Over the subsequent interval $[t, t + K)$, $z_{i,t}$ remains fixed and is passed to the low-level controller as a persistent high-level command. This “commit-and-execute” mechanism reduces high-level decision oscillation and provides temporal consistency for tactical behaviors. In the current framework, teammates’ intentions are neither explicitly communicated nor estimated through a separate belief module during execution.

Cooperating with the Manager is the low-level maneuver execution network, whose policy distribution is parameterized by π_θ . The Worker is responsible for high-frequency flight control and immediate maneuver response, and directly outputs the low-level action $a_{i,t}$. In terms of input design, the Worker uses not only the real-time observation $o_{i,t}$ at the current moment, but also explicitly concatenates the intention vector $z_{i,t}$ generated by the Manager as conditional information, so that action selection becomes an intention-conditioned decision process. The Worker also adopts a GRU to maintain short-term maneuver memory, such as the encoding of local dynamic trends. Its hidden-state update and action generation process are defined as

$$h_{i,t}^{\text{wkr}} = \text{GRU}_{\text{wkr}}([o_{i,t}, z_{i,t}], h_{i,t-1}^{\text{wkr}}), \quad a_{i,t} \sim \pi_\theta(\cdot | h_{i,t}^{\text{wkr}}). \quad (16)$$

Through the above hierarchical temporal design, the Worker actually learns a family of sub-policies conditioned on intentions. That is, given $z_{i,t}$, the complex mapping from observation to maneuver is restricted to a semantically structured subspace. In this way, the originally coupled problems of macroscopic tactical planning and microscopic action execution are decomposed into two levels of learning tasks, thereby improving learnability and behavioral coherence in complex air-combat scenarios.

5.2. Intention-aware counterfactual value evaluation

One of the core difficulties in multi-UAV cooperative control lies in credit assignment. When the team reward is jointly produced by the combined behaviors of multiple agents, how to effectively attribute the global return to the action choice of each individual directly affects the variance of policy-gradient estimation and the convergence quality. To this end, AICC constructs a centralized Critic network $Q_\psi(s, \mathbf{a}, \mathbf{z})$ during training for centralized evaluation of the joint policy. Different from conventional multi-agent methods, the input of this Critic is further extended from the joint action $\mathbf{a}$ and global state s to the joint intention $\mathbf{z} = \{z_1, \dots, z_N\}$. Since $\mathbf{z}$ encodes the high-level intentions generated by the Manager networks, introducing $\mathbf{z}$ enables the Critic to distinguish different coordination modes under the same physical state, thereby improving intention-conditioned value estimation during centralized training.

The Critic network is updated by minimizing the temporal-difference error, with the loss function

$$\mathcal{L}(\psi) = \mathbb{E}_{\tau \sim \mathcal{D}} \left[(y_t - Q_\psi(s_t, \mathbf{a}_t, \mathbf{z}_t))^2 \right], \quad (17)$$

where the target value is

$$y_t = r_t^{\text{team}} + \gamma Q_\psi(s_{t+1}, \mathbf{a}_{t+1}, \mathbf{z}_{t+1}).$$

Here, $r_t^{\text{team}} = \sum_{i \in \Omega_b} r_{i,t}$ denotes the globally shared team reward of the blue-side UAVs at time t , which means that the Critic learns a value signal consistent with overall team performance.

On this basis, to derive the individual policy gradient for UAV i and reduce the gradient noise caused by fluctuations in teammates' actions, a counterfactual baseline technique is adopted. The key idea is to marginalize over the action a_i of UAV i while fixing the actions of the other friendly UAVs, $\mathbf{a}_{-i}$, and the joint intention $\mathbf{z}$, so as to construct an expected baseline consistent with the current policy:

$$b(s_t, \mathbf{a}_{-i,t}, \mathbf{z}_t) = \sum_{a'_i \in \mathcal{A}_{\text{low}}} \pi_\theta(a'_i | h_{i,t}^{\text{wkr}}) Q_\psi(s_t, (a'_i, \mathbf{a}_{-i,t}), \mathbf{z}_t). \quad (18)$$

Accordingly, the advantage function is obtained as

$$A_i(s_t, \mathbf{a}_t, \mathbf{z}_t) = Q_\psi(s_t, \mathbf{a}_t, \mathbf{z}_t) - b(s_t, \mathbf{a}_{-i,t}, \mathbf{z}_t). \quad (19)$$

It should be emphasized that the hidden state $h_{i,t}^{\text{wkr}}$ encodes the local observation $o_{i,t}$ together with the current latent intention $z_{i,t}$. Therefore, the baseline term uses $\pi_\theta(a'_i | h_{i,t}^{\text{wkr}})$ to weight actions, ensuring that the counterfactual expectation is consistent with the current intention-conditioned policy distribution. This advantage function measures the marginal contribution of the specific action $a_{i,t}$ relative to the average action performance under the same intention configuration, thereby providing lower-variance and more informative gradient signals for policy updates.

5.3. Intention-consistency supervision and optimization objective

Although the hierarchical structure explicitly provides the intention z_i as a conditional input to the Worker, the Worker may still gradually weaken or even ignore this conditional information during optimization if no additional constraint is imposed. As a result, the intention-action mapping may degenerate into a policy independent of z_i , causing the high-level intention to lose its actual control significance. To ensure that the generated intentions possess executable tactical semantics and can be inferred by teammates through observation during coordination, a consistency supervision mechanism is introduced to regularize the policy distribution of the Worker.

Based on the intention prior distribution $\rho(a | s, z)$, a regularization term is added to the Worker loss by minimizing the KL divergence between the learned policy distribution and the prior distribution, so that under a given state and intention, the action distribution of the Worker remains consistent with the prior:

$$\mathcal{L}_{\text{consist}}(\theta) = \mathbb{E}_{\tau \sim \mathcal{D}, z \sim \mu} \left[D_{\text{KL}}(\pi_\theta(\cdot | h_{i,t}^{\text{wkr}}) \| \rho(\cdot | s_{i,t}, z_{i,t})) \right]. \quad (20)$$

In terms of the training objective, AICC improves the policy through the counterfactual advantage function while constraining the deviation of the intention-conditioned action distribution through the consistency term. The final optimization objective can thus be

written as maximizing the counterfactual advantage while minimizing the consistency error, and the corresponding gradient update is

$$\nabla_{\theta} J = \mathbb{E}_{\mathbf{x} \sim D} \left[A_i(s_t, \mathbf{a}_t, \mathbf{z}_t) \nabla_{\theta} \log \pi_{\theta}(a_{i,t} | h_{i,t}^{\text{wkr}}) \right] - \lambda \nabla_{\theta} \mathcal{L}_{\text{consist}}(\theta). \quad (21)$$

Here, the first term performs low-variance gradient-ascent updates based on the counterfactual advantage, while the second term controls the strength of the consistency constraint through the coefficient λ .

6. Experimental validation and analysis

In this section, the training results of the AICC framework are analyzed in depth, and the proposed decision-making method is evaluated in a self-developed air-combat simulation environment. In addition, ablation experiments are conducted to quantitatively assess the contribution of each core component to the final cooperative performance.

6.1. Experimental setup

To verify the effectiveness of the AICC algorithm in multi-UAV cooperative air combat, a lightweight multi-agent air-combat simulation platform was developed in Python. Two representative confrontation scenarios were designed: (1) a symmetric 2v2 cooperative scenario, in which both sides each deploy two UAVs with symmetric initial positions and states; and (2) an asymmetric 2v3 disadvantage scenario, in which two friendly UAVs confront three enemy UAVs. This scenario simulates high-difficulty operations under numerical inferiority.

It is assumed that the UAVs on both sides have identical performance. The parameter settings follow Zhang et al. [27]. Specifically, the initial flight speed is set to 200 m/s, the maximum detection range is 6000 m, the maximum pointing angle is 45°, the maximum attack range is 1000 m, the maximum attack angle is 60°, the minimum available tangential overload is 1, the maximum available tangential overload is 2, the minimum available normal overload is 0, the maximum available normal overload is 8, the minimum flight speed is 90 m/s, and the maximum flight speed is 400 m/s. The air-combat episode terminates once either side accomplishes its tactical objective.

In our simulation environments, the red-side opponent UAVs are controlled by a fixed rule-based heuristic system. During both training and evaluation, the opponent's strategy is deterministic given the same state and does not involve network parameter updates. This design provides a consistent and reproducible benchmark for verifying whether the blue-side MARL algorithms can stably converge and learn complex cooperative tactics.

To comprehensively evaluate the performance advantages of AICC, the proposed method is compared with several advanced deep reinforcement learning algorithms:

- **QMIX** [23]: a value-based reinforcement learning algorithm for multi-agent cooperation. It factorizes the centralized action-value function into a monotonic mixing of per-agent utilities via a mixing network.
- **TD3** [28]: a deterministic policy-gradient algorithm based on the Actor–Critic architecture. By introducing twin Critic networks and a delayed update mechanism, it effectively alleviates the problem of Q-value overestimation.
- **MADDPG** [29]: a multi-agent deep deterministic policy gradient algorithm that adopts CTDE architecture.
- **COMA** [30]: a reinforcement learning algorithm for multi-agent cooperation, which reduces the variance of policy gradients by introducing a counterfactual baseline, thereby improving the learning performance of decentralized agents.

The Manager's decision interval is set to $K = 5$, i.e., high-level tactical planning is performed every 0.5 s (with a simulation step size of 0.1 s). Rather than claiming this value is mathematically optimal without exhaustive ablation, $K = 5$ is selected empirically as a domain-informed heuristic. In aerial dogfighting, a 0.5 s macro-action duration is consistent with UAV kinematic inertia and the OODA loop constraints, and it helps prevent physically unrealistic high-frequency jitter in tactical intentions.

In our implementation, the action space cardinality is fixed to $|A_{\text{low}}| = M = 9$. To quantify the measured computational cost, we profile the wall-clock time of a single gradient update step for AICC and a standard MADDPG baseline on the same hardware and with the same training settings. The measured training time per gradient step is $T_{\text{MADDPG}} = 18.4 \text{ ms/step}$ for MADDPG and $T_{\text{AICC}} = 24.7 \text{ ms/step}$ for AICC, resulting in an additional overhead of $(T_{\text{AICC}} - T_{\text{MADDPG}}) / T_{\text{MADDPG}} \times 100\% = 34.2\%$.

All experiments are conducted on a server running Ubuntu 18.04.6 LTS. The server is equipped with a 12th Gen Intel(R) Core(TM) i7-12700K processor, an NVIDIA GeForce RTX 3090 GPU, and 32 GB of RAM. The software environment is built on Python 3.8 and PyTorch 1.12.

6.2. Simulation results and analysis

6.2.1. 2v2 air-combat results

To verify the learning efficiency and asymptotic performance of the proposed algorithm in multi-UAV cooperative air combat, AICC was compared with three mainstream reinforcement learning baselines under the symmetric 2v2 confrontation scenario. Fig. 5 reports the learning curves of average cumulative episode reward versus training episodes in the 2v2 symmetric scenario. The solid lines are the mean over 10 random seeds, and shaded bands indicate standard deviation, respectively.

TD3 rapidly improves in the early stage, but its performance becomes unstable once coordination requirements increase. Because TD3 lacks a centralized perspective and an explicit coordination mechanism, each agent faces an effectively non-stationary learning

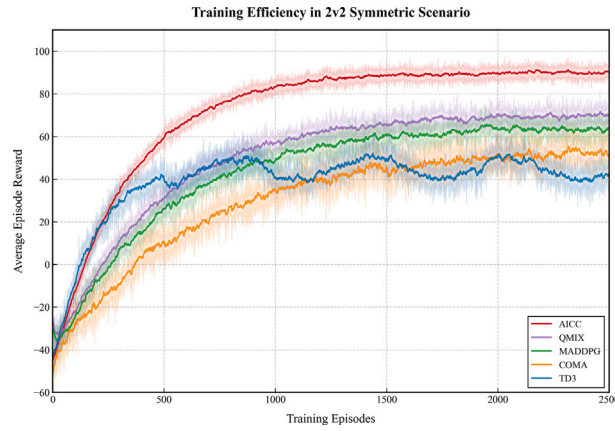


Fig. 5. Training curves of different algorithms in the symmetric 2v2 air-combat scenario.

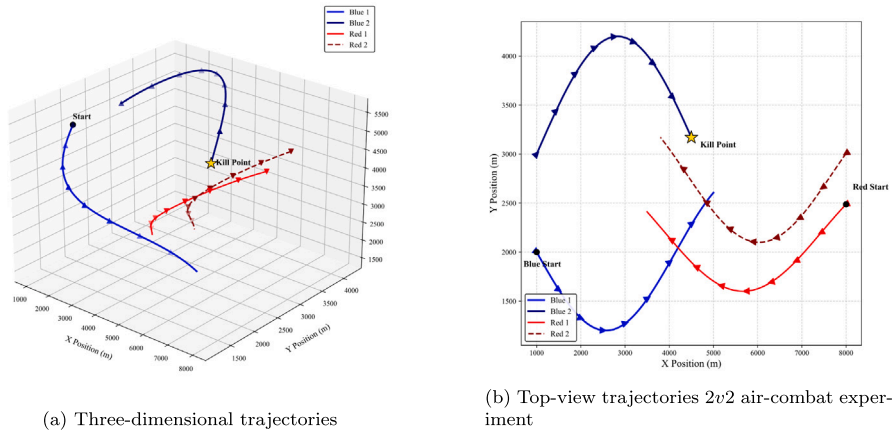


Fig. 6. 2v2 air-combat experiment.

environment, leading to oscillatory dynamics and a low final plateau. In contrast, MADDPG and COMA mitigate non-stationarity via centralized training with decentralized execution. MADDPG shows a more stable trend, yet converges to a suboptimal asymptote, consistent with the limited exploration efficiency induced by the flattened state-action mapping over the joint continuous action space. COMA improves stability relative to TD3 but achieves inferior final performance, with most of its convergence in the range of 45–55, largely due to higher variance in policy-gradient-based credit assignment under continuous actions.

QMIX converges faster than MADDPG in the mid stage and attains a higher plateau, but exhibits larger post-convergence fluctuations than AICC. AICC achieves the best and most stable learning behavior, reaching the highest asymptotic region after roughly 1200 episodes and maintaining low variance thereafter. This is attributed to the hierarchical decomposition together with the intention-consistency objective and counterfactual credit assignment, which jointly provide both credible signaling and properly attributed optimization gradients.

As shown in Fig. 6, both the red and blue formations initially approach each other head-on. The red side explores in the original continuous action space by maximizing immediate rewards, with the two UAVs advancing in a tight formation directly toward the blue side. In contrast, the two blue UAVs do not adopt a conventional frontal engagement strategy. Blue 1 actively reduces altitude and cuts into the right flank, while Blue 2 maintains a large pitch-up climb and executes a wide high-altitude detour toward the left flank. In the middle stage of engagement, the air combat evolves into a typical close-range dogfight. Blue 1 performs aggressive roll maneuvers and sustained high-overload circling, successfully drawing the full attention of the two red UAVs. They are drawn into pursuing Blue 1 into a low-altitude energy-depletion battle. Although they obtain a slight local rear-aspect angular advantage, this comes at the cost of rapidly consuming both altitude potential energy and speed kinetic energy, and their radar field of view and weapon envelope completely lose coverage of the high-altitude airspace. When the red side is exhausted in terms of energy state, Blue 2 dives into the rear hemisphere of the red formation from the high-altitude blind zone at extremely high entry speed, completing weapon locking and kill execution before the red side has any time to react.

We further conducted 200 independent evaluation episodes with random initialization under the symmetric 2v2 scenario. The outcome criteria were defined as follows: if all enemy UAVs are destroyed within the maximum simulation steps while at least one friendly UAV survives, the result is counted as a *win*; if all friendly UAVs are destroyed, the result is counted as a *loss*; if both sides

Table 1
Statistics of win rates over 200 test episodes in the 2v2 air-combat scenario.

Algorithm	Win Rate (%)	Draw Rate (%)	Lose Rate (%)
AICC	81.50	6.50	12.00
MADDPG	60.50	15.00	24.50
COMA	55.00	13.00	32.00
TD3	43.50	16.00	40.50
QMIX	73.00	12.50	14.50

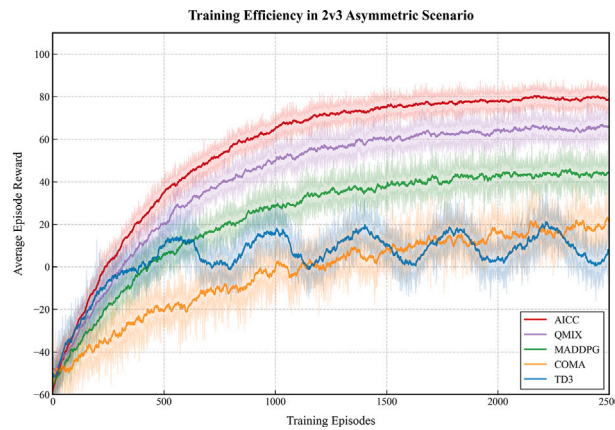


Fig. 7. Training curves of different algorithms in the asymmetric 2v3 air-combat scenario.

still have surviving UAVs when the time limit is reached, or if mutual destruction occurs, the result is counted as a *draw*. The detailed statistics are reported in Table 1.

Table 1 reports the win rate, draw rate, and loss rate over 200 independent evaluation runs for each algorithm. Overall, AICC achieves the most favorable outcome distribution: it attains the highest win rate (81.50%) and the lowest loss rate (12.00%), together with a relatively small draw rate (6.50%). Since win/draw/loss are complementary under the same evaluation protocol, this profile simultaneously reflects stronger capability to secure victories, fewer decisive failures, and a lower frequency of indecisive outcomes, thereby indicating more reliable and stable cooperative behavior during evaluation.

Beyond the point estimates, the statistical confidence of the observed superiority is further assessed. Under a binomial normal approximation with $n = 200$, AICC's win rate of 81.5% yields an approximate 95% confidence interval of [76.1%, 86.9%], while the corresponding intervals for MADDPG, COMA, and TD3 are [53.7%, 67.3%], [48.1%, 61.9%], and [36.6%, 50.4%], respectively. The separation between these intervals suggests that AICC's higher win rate is unlikely to be explained by evaluation noise. Similarly, for loss outcomes, AICC's loss rate of 12.0% corresponds to an approximate 95% confidence interval of [7.5%, 16.5%], which is well below those of MADDPG [18.5%, 30.5%], COMA [25.5%, 38.5%], and TD3 [33.7%, 47.3%]. This reinforces that AICC reduces catastrophic failure events in a statistically meaningful manner rather than only achieving higher average rewards by chance.

In comparison with non-hierarchical baselines, AICC maintains a consistent advantage in both success and failure modes. Specifically, MADDPG, COMA, and TD3 all exhibit lower win rates and higher loss rates than AICC, while their draw rates remain relatively non-negligible, implying more frequent ambiguous or unstable coordination outcomes. This pattern is consistent with the difficulty of learning robust long-horizon coordination under strong multi-agent pressure using flattened actor-critic structures without explicit intention/communication mechanisms.

For stronger hierarchical baselines, QMIX provides improved robustness relative to MADDPG/COMA/TD3, but AICC remains superior in the win-loss trade-off. AICC achieves the highest win rate and the lowest loss rate among all compared methods, while also retaining a smaller draw rate. From a behavioral perspective, the consistently reduced loss rate implies fewer instances of severe mis-coordination; therefore, the learned coordination strategy is more resilient under uncertainty and yields more stable team-level performance across evaluation runs.

6.2.2. 2v3 air-combat results

To further investigate the performance and robustness of different algorithms under extremely complex and high-pressure conditions, the test scenario was upgraded to an asymmetric 2v3 disadvantage confrontation setting. In this scenario, our side faces a numerical disadvantage. Fig. 7 shows the training curve comparison among different algorithms in the 2v3 asymmetric scenario. The solid lines denote the mean over 10 random seeds, and the shaded regions represent the standard deviation.

As the environment becomes more challenging, the baseline methods generally suffer from clear performance degradation. TD3 suffers from severe performance degradation: after a short rebound around episodes, its learning collapses and then oscillates severely within a low-reward band, indicating that single-agent learning cannot handle the strong multi-enemy non-stationarity without

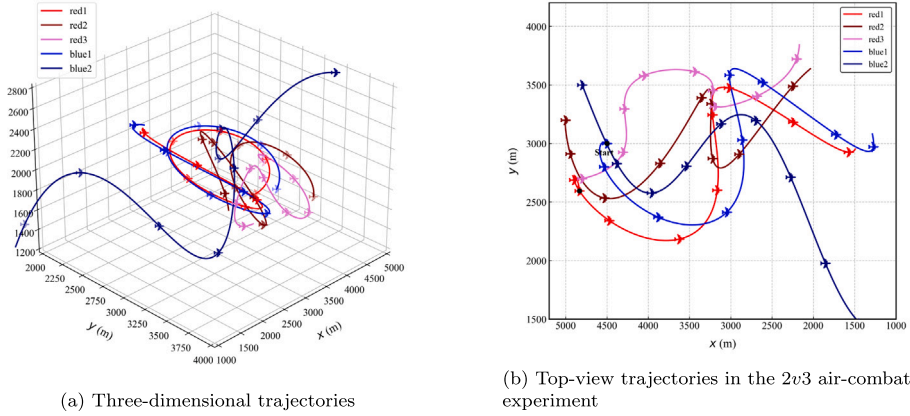


Fig. 8. 2v3 air-combat experiment.

Table 2

Statistics of win rates over 200 test episodes in the 2v3 air-combat scenario.

Algorithm	Win Rate (%)	Draw Rate (%)	Lose Rate (%)
AICC	53.50	15.00	31.50
MADDPG	37.50	14.50	48.00
COMA	23.50	11.00	65.50
TD3	14.00	9.50	76.50
QMIX	48.00	23.50	28.50

intention-level coordination. MADDPG exhibits only a weak positive trend and finally converges around. COMA is even more unstable, with its reward frequently staying below zero in most training stages, which suggests that policy-gradient variance in continuous control amplifies the difficulty under asymmetric pressure. Compared with MADDPG and COMA, QMIX converges to a higher region but still shows noticeable variance and slower stabilization under asymmetric interactions. In contrast, our model maintains a stable, monotonic increase throughout training and converges to a significantly higher reward region after about episodes.

Fig. 8 present the 3D trajectories and top-view trajectories in the 2v3 scenario. At the beginning of the engagement, when facing the frontal pressure launched by the red three-UAV formation with numerical superiority, the blue side does not adopt either a conventional frontal engagement or collective evasion strategy. Instead, it decisively executes an intention-decoupled separation tactic. Blue 1 (light blue curve) is assigned the role of a “decoy” and flies directly into the red-side fire-control zone. Meanwhile, Blue 2 (dark blue curve) receives the intention of “concealed flanking”, greatly expands its lateral separation toward the left flank before contact, and then enters a steep climb, rapidly escaping the frontal radar cone and line-of-sight range of the red side.

In the middle stage of the engagement, Blue 1 exploits the low-level continuous maneuvering capability provided by the Worker network and successfully attracts all three red UAVs into the center of the combat airspace (near $x \approx 3000$ m, $y \approx 2500$ m) through continuous horizontal S-shaped maneuvers and circling. Since the independently learned red-side agents lack global coordination constraints, they fall into a severe “target fixation” trap. The three red UAVs crowd together and are forced to follow Blue 1 into an energy-consuming central dogfight, thereby completely losing situational awareness over the high-altitude and flanking airspace.

In the late stage of the battle, Blue 2, which has remained on the periphery of the main battlefield, demonstrates a high level of tactical sophistication. Through the earlier concealed climb, Blue 2 secures an absolute altitude advantage of as much as 2800 m. Once the three red UAVs become exhausted in both speed and energy due to sustained circling and their formation falls into severe disorder, Blue 2 decisively launches a pull-up–dive attack. It rapidly converts the accumulated potential energy into kinetic energy, enters the battlefield at high speed from the upper rear blind zone of the red formation, and executes an effective attack against an unprepared target.

Table 2 summarizes the win rate, draw rate, and loss rate across 200 independent evaluation runs under the 2v3 disadvantage setting. Overall, AICC stands out with the strongest outcome profile: it reaches the highest win rate while keeping the loss rate at the lowest level among all methods. At the same time, its draw rate is not dominant, suggesting that AICC not only achieves favorable results but also avoids getting stuck in indecisive outcomes when the agents face a clear numerical imbalance. The baselines exhibit substantial performance degradation under this hostile configuration. TD3 shows the weakest decisive outcomes, achieving the lowest win rate and the highest loss rate, which implies that it fails to maintain effective team behavior across a large fraction of evaluation runs when opponent advantage is increased. COMA likewise performs poorly, with a win rate of 23.50% and a loss rate of 65.50%. This outcome is consistent with the known challenge of producing reliable coordination in continuous-control multi-agent settings, where credit assignment uncertainty and non-stationary interactions can destabilize the emergent cooperative policy. Compared with TD3 and COMA, MADDPG demonstrates improved resilience but remains inferior to AICC. MADDPG attains a win rate of 37.50% and a loss rate of 48.00%, while its draw rate is close to that of AICC. Taken together, these numbers suggest that MADDPG may occasionally recover under disadvantage, yet its learned coordination is not sufficiently consistent to prevent decisive defeat throughout the 2v3

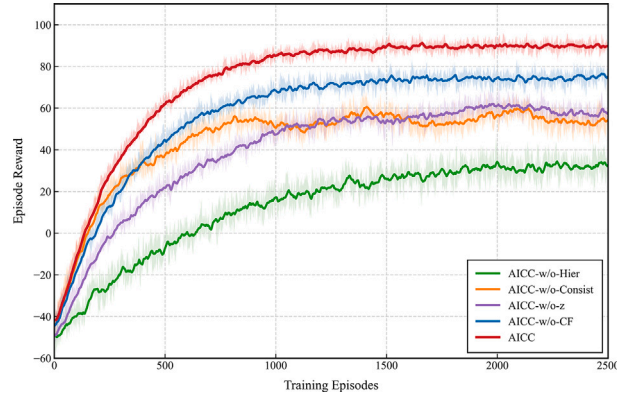


Fig. 9. Comparison of reward curves in the ablation study.

Table 3
Ablation results.

Variants	2v2 Win Rate (%)	2v2 Draw Rate (%)	2v3 Win Rate (%)	2v3 Draw Rate (%)
AICC	81.50	6.50	53.50	15.00
AICC-w/o-CF	73.50	15.00	49.00	22.00
AICC-w/o-Consist	63.50	17.00	42.00	26.00
AICC-w/o-Hier	52.00	19.50	33.00	25.50
AICC-w/o-z	69.50	12.00	47.50	18.50

pressure regime. For the hierarchical baselines, QMIX offers intermediate improvements relative to MADDPG/COMA/TD3. QMIX reaches a win rate of 48.00%, but it also exhibits the largest draw rate along with a moderate loss rate. This pattern suggests that QMIX may more frequently steer the trajectories toward less decisive outcomes, whereas AICC more reliably converts coordinated behavior into wins. Nevertheless, AICC still yields a higher win rate and a lower loss rate, indicating superior effectiveness in realizing coordinated intentions as successful joint execution under asymmetric threat. Under the 2v3 disadvantage scenario, AICC provides the best trade-off between success and failure: it achieves the highest win rate, the lowest loss rate, and a moderate draw rate. These results support the conclusion that AICC enables more robust and controllable coordination under asymmetric threat, thereby reducing the likelihood of persistent mis-coordination that typically results in defeat.

The possible causes for the failure of AICC include the following points: (i) intention grounding failure under partial observability, where key target/ally state variables are occluded or ambiguous, leading agents to form intentions that are only valid at the intention level but inconsistent with the true tactical context; (ii) execution breakdown under focus fire, where concentrated opponent pressure disrupts formations and prevents the timely simultaneous execution of complementary roles, which can propagate into further mis-coordination; (iii) role conflict and over-commitment, where agents pursue locally plausible but globally conflicting behaviors, leaving critical regions or mobility channels insufficiently covered.

6.3. Ablation study

To verify the independent contribution of each innovative mechanism in AICC and its theoretical necessity in Markov games, ablation variants were designed by removing core components. Fig. 9 shows the changes in training reward curves after different components are removed. AICC-w/o-Hier (without Hierarchical Architecture) indicates that, in a long-horizon game such as multi-agent air combat, temporal credit assignment cannot be effectively recovered by learning solely from random perturbations at the low-level continuous action layer. AICC-w/o-Consist (without Intent Consistency Constraint) lacks the constraint that keeps the exploration behavior aligned with the intended coordination direction; when the encounter transitions into a complex dogfight stage, the agents' exploration trajectories can drift and the resulting policy gradients interfere with each other. As training proceeds, the system is more likely to settle into a less favorable local equilibrium. AICC-w/o-CF (without Counterfactual Credit Assignment) removes the counterfactual baseline, so the global reward signal is less informative about each UAV's marginal contribution. In contrast, AICC-w/o-z removes the z -component. Its learning curve still improves consistently and stays clearly above the hierarchical- and consistency-ablated variants, but it converges to a noticeably lower reward than the full AICC. This pattern suggests that z mainly supports higher-quality coordination refinement in later training; without z , the team can still discover partially cooperative behaviors, yet the policy cannot fully reach the same level of stable joint execution under sustained adversarial pressure.

The quantitative performance of all variants over 200 test episodes with random initialization is reported in Table 3.

The degradation pattern in Table 3 highlights the distinct contributions of different AICC components to cooperative learning under both symmetric (2v2) and disadvantaged (2v3) settings.

AICC consistently achieves the strongest overall decision performance across the two experimental regimes. In the relatively balanced 2v2 setting, the full model attains the most favorable win-to-draw trade-off. When the task shifts to the more challenging

Table 4
Average action entropy over the first 500 episodes in the 2v3 scenario.

Method	$\overline{H}_{1:500}$	Entropy gap ΔH
AICC	0.71	–
AICC-w/o-Hier	0.52	0.19

2v3 disadvantage, AICC remains best-in-class in terms of win likelihood while also producing the least frequent non-decisive outcomes among the compared variants.

The ablation results further demonstrate that the mechanisms are non-redundant and operate at different stages of the coordination pipeline. Removing the hierarchical component yields the most pronounced degradation in both settings. The corresponding policy experiences a substantial reduction in win performance and a clear increase in draw occurrences. This outcome aligns with the necessity of hierarchical temporal credit assignment for long-horizon tactical reasoning: without temporal abstraction, the learning signal associated with high-level coordination becomes less attributable to earlier decision steps, which in turn weakens effective exploration and undermines stable convergence toward winning strategies.

Eliminating the intention-consistency constraint also impairs coordinated execution, with the impact becoming more evident under 2v3. In this setting, the agents' coordination is less able to resolve engagements into decisive outcomes, leading to a higher rate of non-decisive results and a lower win propensity. These trends suggest that, when multi-agent synchrony is required, maintaining exploration alignment with the intended joint direction is important for reducing policy-gradient interference and avoiding coordinated tactics that fail to achieve the intended end states.

Regarding credit attribution, removing counterfactual credit assignment causes a comparatively milder degradation, yet the resulting policies remain inferior to AICC. The deterioration manifests as reduced win performance and a higher draw tendency, indicating that counterfactual baselines improve the discriminability of individual UAV contributions by mitigating the ambiguity introduced by global rewards. Finally, ablating the z component produces an intermediate effect: win performance remains relatively closer to AICC, while draw behavior worsens modestly. This relative robustness suggests that z primarily strengthens late-stage coordination refinement, improving the stability of execution and the likelihood of converting coordinated interactions into decisive outcomes.

The performance gap between AICC and AICC-w/o-Hier in the 2v3 scenario is attributed to the loss of effective exploration when the hierarchical structure is removed. To provide a quantitative verification, we measure the average action entropy over the first 500 training episodes. At each low-level decision step t , the low-level policy induces an action distribution $\pi(a | s_t)$. The action entropy at step t is computed as

$$H_t = - \sum_a \pi(a | s_t) \log \pi(a | s_t). \quad (22)$$

We use natural logarithms, so the entropy is reported in *nats*. For an episode e with T_e decision steps, we average across time and then average across the first 500 episodes:

$$\overline{H}_{1:500} = \frac{1}{500} \sum_{e=1}^{500} \left(\frac{1}{T_e} \sum_{t=1}^{T_e} H_{t,e} \right). \quad (23)$$

Table 4 reports $\overline{H}_{1:500}$ for AICC and AICC-w/o-Hier, along with the entropy gap defined as $\Delta H = \overline{H}_{1:500}(\text{AICC}) - \overline{H}_{1:500}(\text{variant})$.

As shown in Table 4, AICC maintains a higher action entropy than AICC-w/o-Hier throughout the early training stage. Specifically, the average action entropy of AICC-w/o-Hier is lower by 0.19 nats compared to AICC. This indicates that without the hierarchical structure, the low-level policy becomes less stochastic earlier in training.

6.4. Prior corruption ablation

To evaluate the robustness of AICC to imperfect intention–action priors, we conduct a prior corruption ablation in the 2v3 scenario. In AICC, the prior mapping is defined as a categorical distribution over the low-level maneuver action space conditioned on the state s and the high-level intention z :

$$\rho(\cdot | s, z) \in \Delta(\mathcal{A}_{low}), \quad (24)$$

where $|\mathcal{A}_{low}| = 9$.

Let $\eta \in [0, 1]$ denote the prior corruption rate. We corrupt the original prior distribution by interpolating it with a uniform distribution over the low-level action space:

$$\tilde{\rho}(a | s, z) = (1 - \eta)\rho(a | s, z) + \eta \cdot \frac{1}{|\mathcal{A}_{low}|}, \quad a \in \mathcal{A}_{low}. \quad (25)$$

When $\eta = 0$, the original prior is preserved; when $\eta = 1$, the prior becomes a uniform distribution and contains no action preference.

The corrupted prior $\tilde{\rho}$ is used to replace the original prior ρ in the intention-consistency regularization and the intrinsic consistency reward:

$$\mathcal{L}_{consist}(\theta) = \mathbb{E}_{t \sim D, z \sim \mu} \left[D_{\text{KL}} \left(\pi_{\theta}(\cdot | h_{i,t}^{wkr}) \parallel \tilde{\rho}(\cdot | s_{i,t}, z_{i,t}) \right) \right], \quad (26)$$

Table 5
Prior corruption ablation in the 2v3 scenario.

Corruption Rate η	Avg. Episode Reward	Win Rate (%)	Reward Drop (%)
0.0	79.8	53.50	0.00
0.1	77.4	51.00	3.01
0.3	72.1	47.00	9.65
0.5	65.3	40.50	18.17
0.7	56.7	32.50	28.95
1.0	48.9	28.50	38.72

$$r_{i,t}^{consist} = \log \tilde{\rho}(a_{i,t} | s_{i,t}, z_{i,t}). \quad (27)$$

The relative reward drop is computed with respect to the uncorrupted setting:

$$\text{Drop}(\eta) = \frac{R(0) - R(\eta)}{R(0)} \times 100\%, \quad (28)$$

where $R(0)$ is the average episode reward under the original prior and $R(\eta)$ is the average episode reward under corruption rate η . To reduce evaluation variance and obtain more reliable estimates, we evaluate each corruption level over 200 test episodes. Table 5 reports the average episode reward, win rate, and relative reward drop under different prior corruption rates. The reward drop is computed with respect to the uncorrupted setting:

$$\text{Drop}(\eta) = \frac{R(0) - R(\eta)}{R(0)} \times 100\%, \quad (29)$$

where $R(0)$ denotes the average episode reward without prior corruption and $R(\eta)$ denotes the average episode reward under corruption rate η .

As shown in Table 5, the performance of AICC degrades progressively as the prior corruption rate η increases. When the corruption $\eta = 0.1$, the average episode reward decreases only from 79.8 to 77.4, corresponding to a reward drop of 3.01%, and the win rate decreases slightly from 53.50% to 51.00%. This indicates that AICC is not overly sensitive to small perturbations in the intention prior distribution.

Under moderate corruption, the degradation becomes more visible but remains gradual. When $\eta = 0.3$, AICC still achieves an average reward of 72.1 and a win rate of 47.00%, with a reward drop of 9.65%. This suggests that the learned hierarchical policy can partially compensate for inaccurate prior guidance through its low-level adaptive decision-making.

When the corruption rate becomes large, the performance deterioration is more pronounced. At $\eta = 0.7$, the average reward drops to 56.7 and the win rate decreases to 32.50%; when $\eta = 1.0$, where the prior becomes completely uninformative, the reward further decreases to 48.9 and the win rate to 28.50%. These results show that the intention prior provides meaningful guidance for aligning high-level tactical intentions with low-level maneuver execution.

7. Conclusion

This paper addresses the problem of multi-UAV cooperative air-combat decision-making under incomplete information, strong adversarial confrontation, and partial observability, and proposes the AICC framework for intention-consistent cooperative combat. Under the CTDE paradigm, the decision-making process is decoupled into a hierarchical structure of “high-level intention–low-level maneuver”: the Manager generates discrete tactical intentions at a low frequency, while the Worker executes high-frequency discrete maneuvers conditioned on the current intention. Meanwhile, an intention-consistency supervision mechanism based on the intention prior mapping ρ is introduced, including both KL regularization and intrinsic consistency rewards, so as to suppress coordination mismatches caused by inconsistency between intention and action. In addition, an intention-aware centralized Critic and a counterfactual baseline are constructed to achieve more stable credit assignment. Future work will focus on a more rigorous evaluation of AICC under practical deployment conditions. In particular, we will test the method under realistic uncertainties such as sensor noise, delay, packet loss, and limited communication. In the future, we will no longer rely on fixed time abstractions but instead study adaptive time abstractions, enabling agents to autonomously learn the termination conditions of tactical intentions, thereby achieving dynamic and scenario-adaptive time abstractions without the need for exhaustive hyperparameter tuning.

CRedit authorship contribution statement

Mingwei Zhao: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Junwu Zhu:** Formal analysis, Conceptualization. **Mingxuan Liang:** Supervision, Software. **Yuanyuan Zhang:** Supervision. **Die Ge:** Project administration. **Xu Liu:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Detailed hyperparameter settings

Table A.6 provides the comprehensive hyperparameter settings used for training and evaluating the AICC framework. This table extends the basic network settings by including the detailed exploration schedule, entropy coefficient, target network update frequency, and the evaluation protocol.

Table A.6
Hyperparameters of the AICC framework.

Hyperparameter	Value
<i>Network and Optimization</i>	
Actor learning rate	5×10^{-4}
Critic learning rate	1×10^{-3}
Optimizer	Adam
Discount factor (γ)	0.99
Mini-batch size	64
Replay buffer capacity	10^5
GRU hidden state dimensionality	128
Target network update rate (τ)	0.005
<i>Algorithm-Specific and Training</i>	
Total training episodes	2500
Manager intention update interval (K)	5
Intention space cardinality ($ \mathcal{Z} $)	4
Intention-consistency coefficient (λ)	0.1
Intrinsic reward weight (β)	0.05
Shaping reward weights (ω_d, ω_a)	0.5, 0.5
Exploration schedule	ϵ -greedy ($1.0 \rightarrow 0.05$)
Entropy coefficient	0.01
Evaluation protocol	Greedy

Data availability

Data will be made available on request.

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